Paraphraser Fingerprinting: Can We Identify the Rewriter from the Text Alone?

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1 Problem Description

Our research question is:

> **RQ: Paraphraser Fingerprints.** Do different paraphrasing models leave distinct, detectable traces in their outputs? Can we identify the paraphraser from the text alone?

2 Dataset

Each experimental run produces 20 trials. Only the initial trial uses a human-crafted source text containing deliberate linguistic traps (non-standard grammar, intentional repetition, temporal disorder, counting exercises). All subsequent trials use source texts generated by the Auditor LLM using Automatic Prompt Engineering (APE)-style iterative refinement based on past Detective feedback.

3 Experimentation and Results

3.1 Experimental Setup

3.2 Aggregate Accuracy

3.3 Per-Trial Learning Curve

Figure 2 shows the per-trial accuracy averaged across runs. The first 5 trials serve as warmup. The cumulative accuracy curve (red) shows the running average from trial 5 onward, revealing the Detective's performance trajectory as the Auditor iteratively refines its source text generation strategy.

3.4 Per-Paraphraser Detection Rates

Detection rates vary dramatically across paraphrasers (Figure 3):

- **GPT-5-Nano**: 100% (36/36)—always detected when duplicated.
- **Gemini-2.5-Flash**: 64.3% (18/28).
- **DIPPER**: 42.6% (20/47).
- **Llama-3.1-8B**: 25.0% (6/24)—hardest to detect.

3.5 Confusion Analysis

The aggregated confusion matrix (Figure 4) reveals systematic patterns in the Detective's errors:

- **GPT-5-Nano ↔ DIPPER** (off-diagonal: 32): The most common confusion pair. Despite their architectural differences (LLM vs. T5-based), the Detective frequently groups their outputs together.
- **Gemini-2.5-Flash ↔ GPT-5-Nano** (14): Moderate confusion between these two LLM-based paraphrasers.
- **Llama-3.1-8B ↔ Gemini-2.5-Flash** (8): Some confusion between these models.
- **DIPPER ↔ Llama-3.1-8B** (0): Never confused, suggesting maximally distinct fingerprints.

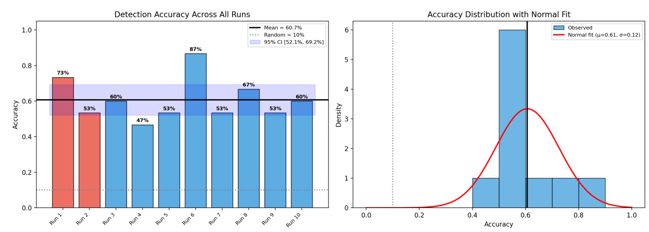

Figure 1: Detection accuracy across all 10 runs. Left: per-run accuracy with mean and 95% CI. Right: accuracy distribution with normal fit. The mean accuracy of 60.7% is significantly above the 10% random baseline ($t(9) = 13.41$, $p < 0.001$).

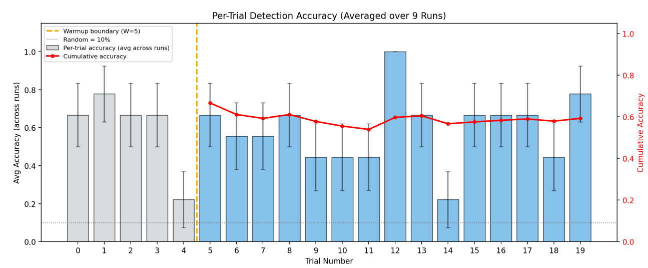

Figure 2: Per-trial detection accuracy averaged over 9 runs with full data. Blue bars show average accuracy at each trial position; the red curve shows cumulative accuracy from trial 5 onward. Error bars indicate standard error.

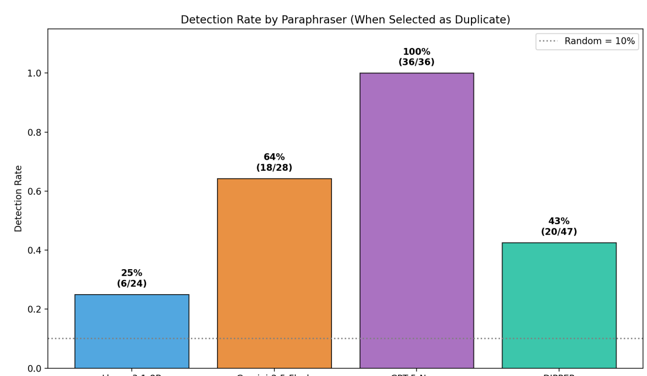

Figure 3: Detection rate for each paraphraser when randomly selected as the duplicate source. GPT-5-Nano is detected with 100% accuracy, while Llama-3.1-8B is hardest to detect at 25%.

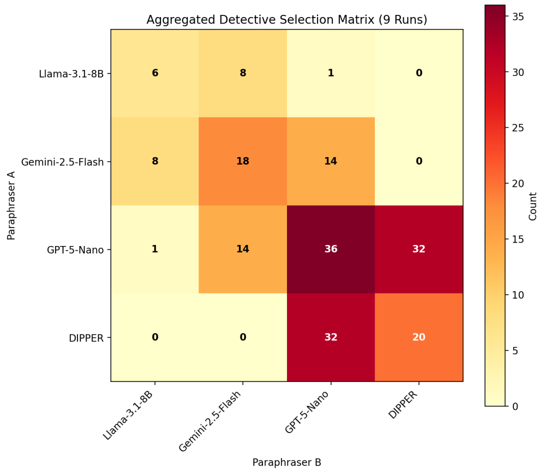

Figure 4: Aggregated confusion matrix showing how often the Detective paired each combination of paraphrasers. Diagonal entries represent correct identifications. The strong GPT-5-Nano–DIPPER off-diagonal (32) indicates the Detective frequently confuses these two.

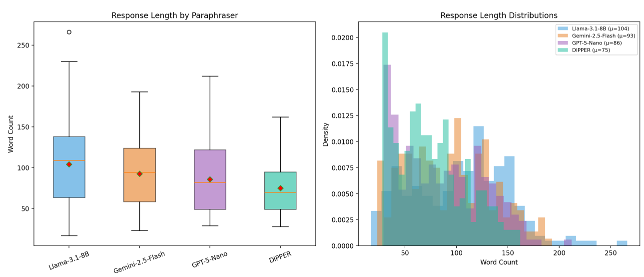

Figure 5: Response length (word count) distributions per paraphraser. Left: box plots. Right: overlaid histograms. Llama-3.1-8B produces the longest outputs ($\mu = 104$), while DIPPER produces the shortest ($\mu = 75$).

Table 1: Response length statistics (word count) per paraphraser.

Paraphraser	Mean	Median	SD	n
Llama-3.1-8B	104	109	47	210
Gemini-2.5-Flash	93	94	41	216
GPT-5-Nano	86	82	41	228
DIPPER	75	70	33	246

3.6 Response Length as a Fingerprint Feature

Response length is a salient fingerprint dimension (Figure 5). Table 1 summarizes the statistics.

DIPPER, as a T5-based model, produces the most conservative (shortest) paraphrases, while Llama-3.1-8B tends toward verbose, elaborative rewrites. Interestingly, despite GPT-5-Nano having moderate length, its fingerprint is the strongest—suggesting that length alone is insufficient and that deeper stylistic features (word choice, sentence structure, formatting patterns) play a critical role.

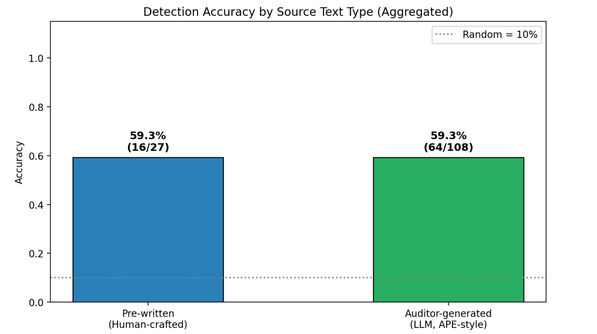

Figure 6: Detection accuracy by source text type. Both human-crafted and Auditor-generated texts yield identical accuracy (59.3%), suggesting the APE-style refinement matches human-designed linguistic traps.

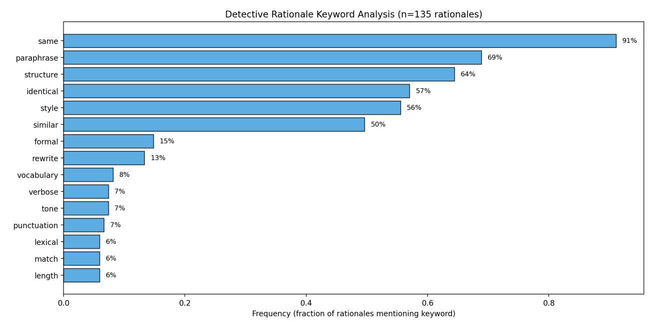

Figure 7: Keyword frequency analysis of Detective rationales (n = 135). Structural and stylistic features dominate the Detective's reasoning.

3.7 Source Text Type Comparison

3.8 Detective Rationale Analysis

Analysis of 135 Detective rationales reveals the features most relied upon for fingerprint identification (Figure 7): *structure* (64%), *style* (56%), *formal* (15%), *vocabulary* (8%), *tone* (7%), and *length* (6%). The dominance of structural and stylistic features over surface-level features like length confirms that the Detective performs genuine stylistic analysis rather than relying on trivial heuristics.

4 Plan for Next Steps

Based on our findings, we propose the following directions:

(1) **Upgrade Detective model.** Replace Paraphrasers or Auditor with a more capable model (e.g., Claude Opus, GPT-5) to test whether stronger reasoning improves fingerprint detection.
(2) **Expand paraphraser pool.** Add more paraphrasers (e.g., Mistral, Command-R, Qwen) to test whether fingerprinting scales to larger model sets and whether models from

the same family (e.g., Llama-3.1-8B vs. Llama-3.1-70B) are distinguishable.

5 Conclusion

- Fingerprint strength varies dramatically across models: GPT-5-Nano is always detectable (100%), while Llama-3.1-8B is rarely detected (25%).
- The Detective relies primarily on structural and stylistic features rather than surface-level heuristics like output length.

These results suggest that the "wood grain" of AI models is real and measurable, opening avenues for paraphraser attribution, AI text provenance, and deeper understanding of how different architectures shape language generation.